# Introduction to Minimum Mean-Squared Error Filtering

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# Key Concepts

1) The generic minimum mean-squared error (MMSE) filtering problem defines an error signal

$$e[n] = d[n] - \sum_{k=0}^{N-1} w_k x[n-k]$$

where $w_k$ are the coefficients of an FIR filter, $d[n]$ and $x[n]$ are input signals.

2) The MMSE design problem chooses $w_n$ to minimize a function of the squared error for a given set of signals $d[n]$ and $x[n]$.

Let $\mathbf{w} = \begin{bmatrix} w_0 & w_1 & \dots & w_{N-1} \end{bmatrix}^T$ be a vector of FIR filter weights.

a) Idealized MMSE

$$\min_{\mathbf{w}} E\{e^2[n]\}$$

b) Practical MMSE

$$\min_{\mathbf{w}} \frac{1}{L} \sum_{n=l_1}^{l_1+L-1} e^2[n]$$

3) This generic MMSE problem formulation may be used to address many important signal-processing applications.

a) For noise/interference suppression we measure $d[n]$ as a noisy signal and $x[n]$ as a signal-free version of the noise.

b) In system modeling or estimation, we let $x[n]$ be the input to the system we seek to model, and $d[n]$ the unknown system's output.

c) In equalization of an unknown system, $x[n]$ is the output of the system producing the distortion and $d[n]$ is the input.

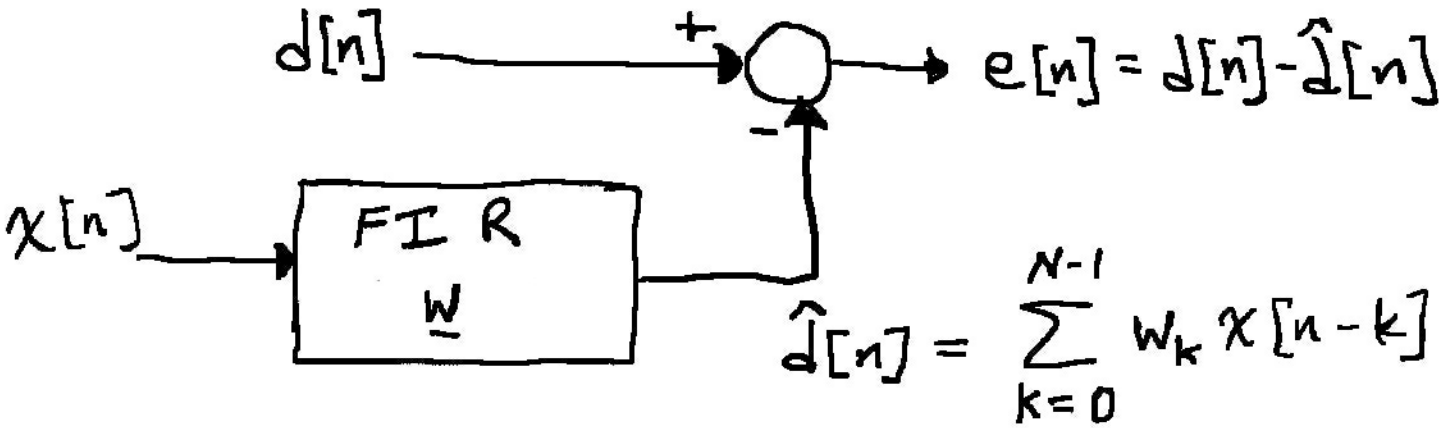
d) In both equalization and modeling known training signals are often employed to design the filter and it is then applied with unknown inputs to the system being modeled or causing distortion.

e) Additional applications include linear prediction and beamforming for sensor (e.g., antenna) arrays.

Minimum mean-squared error (MMSE) filters use measured signals to design filter coefficients

- typically FIR (weighted sum of input signals)
- usually not frequency selective (lowpass, bandpass, etc)
- can adapt to changes in signals

## Generic Formulation



Choose weights $\underline{w}$ to minimize "mean" squared error

$$\underline{w} = [w_0 \; w_1 \; \cdots \; w_{N-1}]^T$$ vector of FIR filter weights

Idealized MMSE

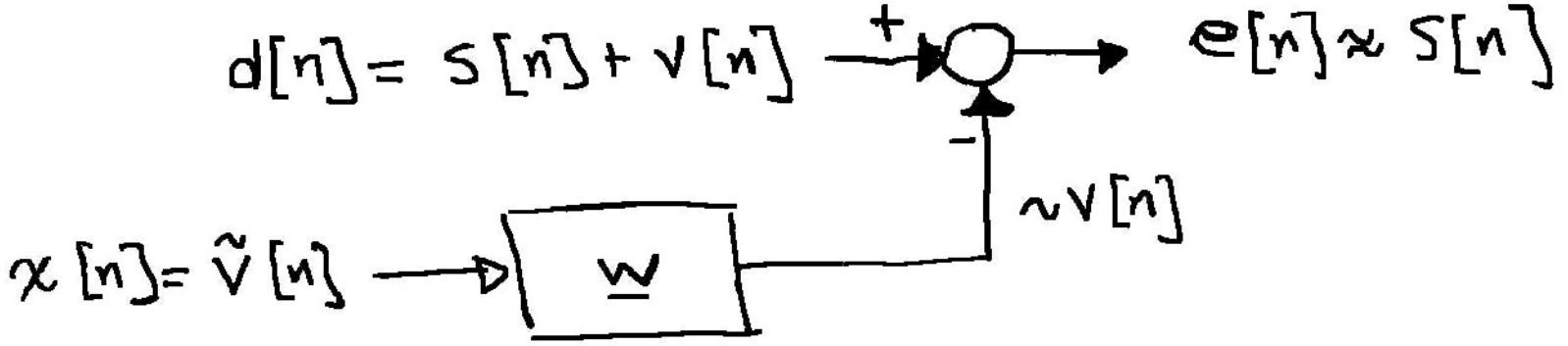
$$\min_{\underline{w}} E\{e^2[n]\}$$ "average" of $e^2[n]$

Practical

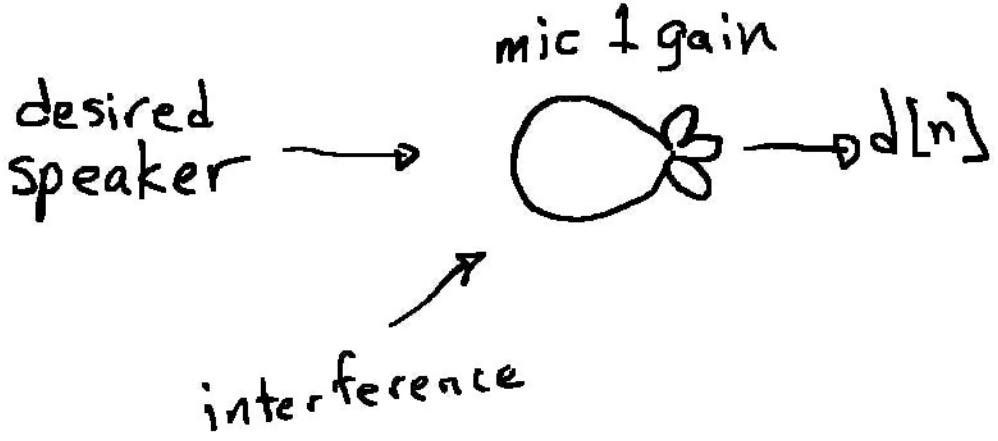
$$\min_{\underline{w}} \frac{1}{L} \sum_{n=l_1}^{l_1+L-1} e^2[n]$$ average of $e^2[n]$ over $L$ time samples

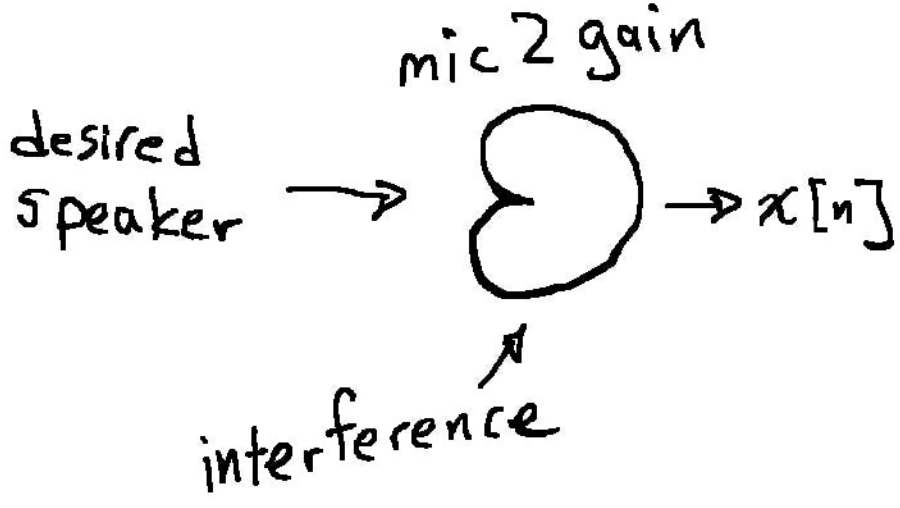
# Application: Noise Suppression






- requires "signal free" noise/interference signal $x[n]$
- signal and $x[n]$ must be "uncorrelated"



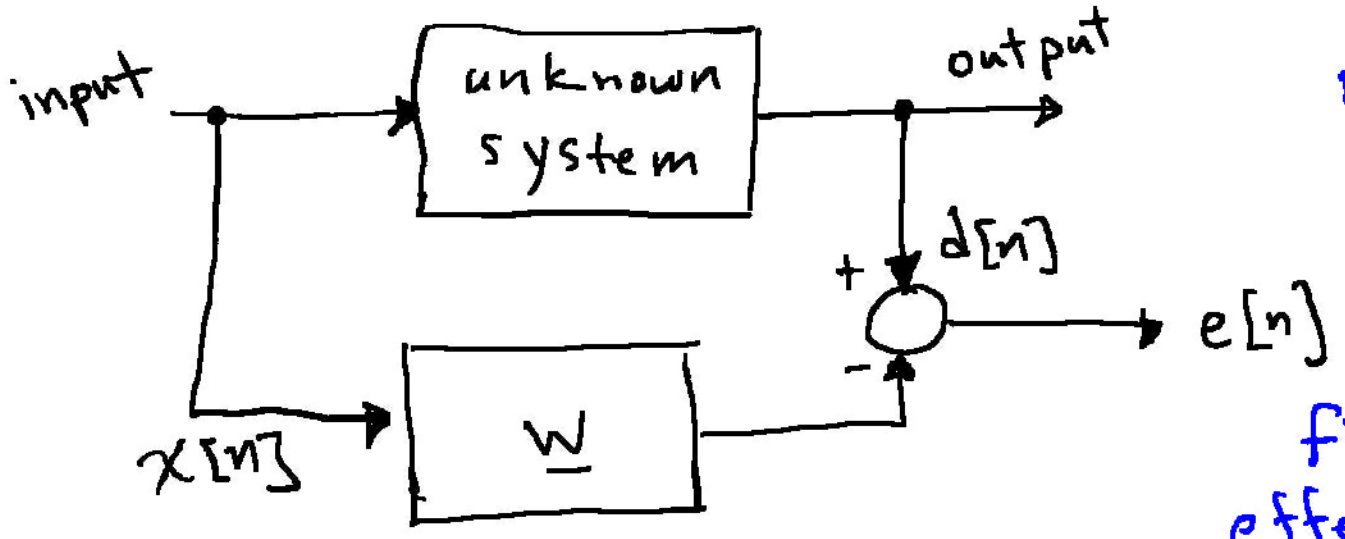



# Application: System Modeling/Estimation



communication channel

physical device
e.g. auditorium

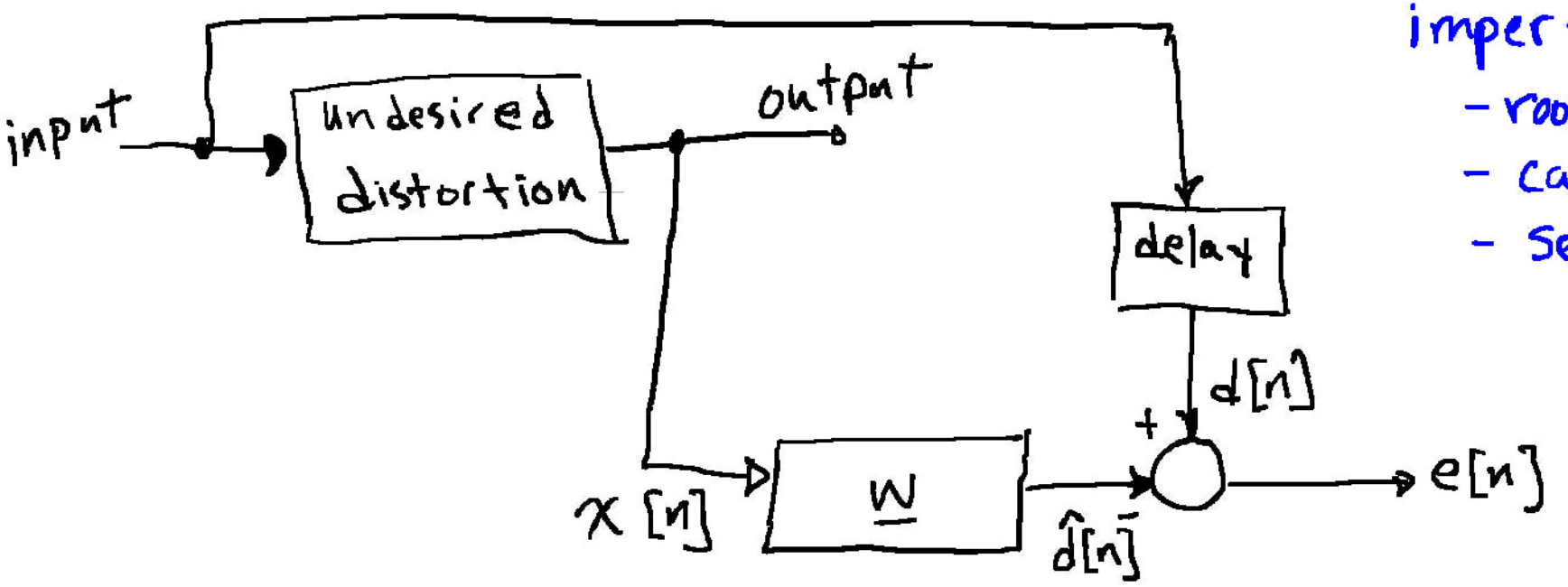



filter $\underline{w}$ models the effect of the unknown system

- Assumes unknown system is linear
- Requires access to system input and output
- "Complete" model requires $x[n]$ contain all frequencies

# Application: Equalization



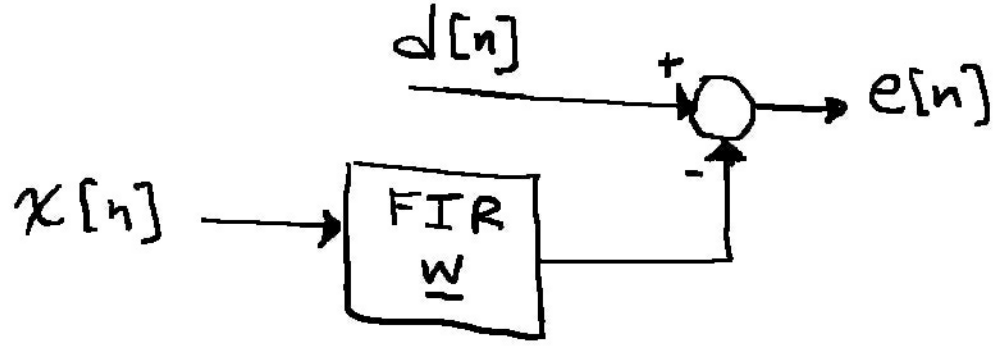



imperfect measurement
- room acoustics
- camera motion blur
- sensor distortion

$\hat{d}[n]$ approximates the delayed input-predistortion

- need access to input and output for designing $\underline{W}$
- delay is needed to account for net timeshift in distortion and in $\underline{W}$
- need sufficiently rich input
- can't equalize or invert zero

# Summary




$$\min_{\underline{w}} E\{e^2[n]\} \Rightarrow \min_{\underline{w}} \frac{1}{L} \sum_{n=\ell_1}^{\ell_1+L-1} e^2[n]$$

MMSE filter formulation applies to many signal-processing problems—

interference suppression, system identification, equalization, linear prediction, beamforming (e.g. antenna arrays), ...

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